Inequalities — problem 7.44

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Let $x_i > 0$, $s = x_1 + \cdots + x_n$. Show that

$$\frac{s}{s-x_1} + \frac{s}{s-x_2} + \cdots + \frac{s}{s-x_n} \geq \frac{n^2}{n-1}.$$